Week 4 report of Internship

Name	Ammar Akram
ID	DVS-8041
Domain	Machine Learning
Ofiice	Devsphere
Date	28/09/2026

DEVSPHERE AI & ML INTERNSHIP

WEEK 4 — CLASSIFICATION MODELS

Breast Cancer Classification using Logistic Regression

1. Project Overview:

This project was completed as part of the DevSphere AI & ML Internship Week 4 task. The objective was to implement a classification model, train it on a classification dataset, generate predictions, and evaluate the model using an accuracy score and confusion matrix.

2. Dataset:

The Breast Cancer Wisconsin dataset available through Scikit-learn was used for this project.

Dataset Detail	Value
Total Samples	569
Total Features	30
Classes	Malignant, Benign
Training Samples	455
Testing Samples	114

3. Data Preparation:

The dataset was divided into training and testing sets using an 80/20 split.

The following settings were used:

- Test size: 20%
- Random state: 42
- Stratified train/test split
- Feature scaling using StandardScaler
- Feature scaling was applied before training the classification model to standardize the input features.

4. Classification Model:

The classification algorithm used for this task was Logistic Regression.

The model was trained using the scaled training dataset and then used to predict the class of each sample in the testing dataset.

The two classification classes were:

- Malignant
- Benign

5. Predictions:

After training, the model successfully generated predictions for the 114 testing samples.

Sample predictions:

[0, 1, 0, 1, 0, 1, 1, 0, 0, 0]

6. Model Evaluation:

The trained model achieved an accuracy of:

98.25%

Evaluation Metric	**Result**
Model	Logistic Regression
Test Samples	114
Accuracy	98.25%

The classification report also showed strong precision, recall, and F1-score for both classification classes.

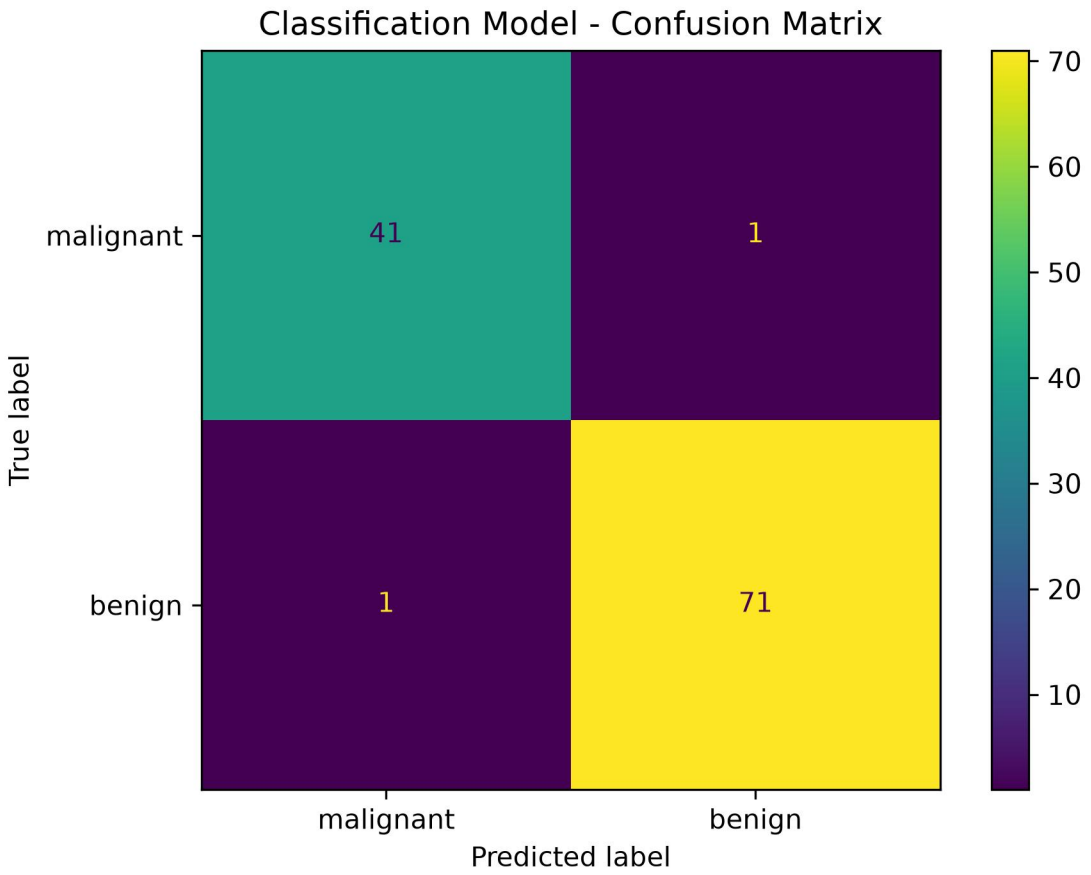
7. Confusion Matrix:

The confusion matrix was generated to compare the actual classifications with the predictions made by the model.

Confusion Matrix

[[41, 1],

[1, 71]]



8. Model Saving:

The trained classification model was saved as:

classification_model.pkl

The saved file contains the trained Logistic Regression model, feature scaler, feature names, and target class names.

9. Technologies Used

- Python
- Pandas
- Scikit-learn
- Matplotlib
- Joblib
- Jupyter Notebook

11. Conclusion

The Week 4 classification task was successfully completed. A Logistic Regression model was trained using the Breast Cancer Wisconsin dataset and achieved 98.25% accuracy on the test dataset.